

Tanuj Kumar Verma

Junior Software Engineer

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9882798088

Experienced IT Professional with 2.6 years of Experience in the Industry, Specializing in Python for Diverse Software Development, Including Machine Learning and generative AI projects such as CV Parser Tool (Helping HR Recruiters to Select Candidates) including usage of large Language Models (LLM's). Proven ability to analyze and Troubleshoot Complex Technical Issues Promptly. Proficient in Python, MySQL, Exploratory Data Analysis, Machine Learning, RAG, Artificial Intelligence, LlamaIndex, Langchain and Data Visualization. Passionate about Innovation and Continuously Exploring New Software Development Techniques to drive Impactful Business Solutions.

Core Skills

Generative AI - Langchain, LlamaIndex, Large Language Models, Small Language Models, RAG (Retrieval Augmented generation),
Microsoft Azure - Azure AI Search, Azure OpenAI, Microsoft Azure Services,
Languages - Python, SQL,
Operation System - Windows,
Machine Learning, Statistics, Exploratory Data Analysis, Data Visualization, Numpy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook, MySQL Workbench, SQL - Data Query and Manipulation, EDA - Data Cleaning, Outlier Treatment, Scaling, Transformation, Encoding, Statistics - Probability Distribution, Hypothesis Testing,
Machine Learning - Supervised and Unsupervised Learning, Regression and Classification,
Cloud Services - WinSCP, PuTTY, Pageant, EC2 Server Deployment, GitLab, VS Code

Education

Great lakes Institute of Management

Dec 2020 - Dec 2021

Post-Graduation Diploma

Data Science and Engineering

DAV College, jalandhar

Jul 2017 - Oct 2020

Bachelor of Science Computer Science

Work Experience

Junior Software Engineer

Jul 2024 - Present

Centelon | Trivandrum

Project Title: Microsoft Azure AI Search

Technology: Microsoft Azure Services, Azure AI Search, Azure OpenAI
Description: I am developing a platform using Microsoft Azure AI Search to interact with documents stored in Azure Blob Storage and SharePoint. The platform creates indexes and uses skillsets to enhance response accuracy. It integrates with Azure OpenAI for query processing, providing a chat playground for user interaction. A web application has been deployed for organizational users to test the combined functionalities of Azure AI Search and OpenAI.

Junior Software Engineer

May 2024 - Jun 2024

Centelon | Trivandrum

Project Title: Small Language Models

Technology: Small Language Models (SLM), RAG (Retrieval Augmented Generation), CTransformers, LlamaCPP, Query Engines.
Description: Conducted Extensive Research on Small Language Models that can run CPUs without requiring GPUs. Developed a RAG Application in order to analyze accuracy and the period of time acquired by disparate Small Language Models to give responses while running on CPUs rather than GPUs.

Junior Software Engineer

Nov 2023 - May 2024

Centelon | Trivandrum

Project Title: CV Parser Tool (Generative AI)

Technology: Langchain, LlamaIndex, Large Language Models, WinSCP, PuTTY, Pageant, Apache, NginX, EC2 Server, Python, RAG (Retrieval Augmented Generation), HuggingFaceHub, OpenAI, MySQL, GitLab, Azure Services.

Description: Developed an online CV processing system utilizing Azure services and machine learning models for efficient document analysis and user query handling. Implemented an offline CV evaluation system using Streamlit, integrating advanced ML algorithms to process multiple CVs concurrently for comprehensive applicant analysis and automated

decision-making. Designed a user-friendly interface for efficient management of applicant data and streamlined evaluation workflows. Deployed the CV parser model on an EC2 server via SCP, integrating GitLab for version control and secure code management. Utilized CI/CD pipelines between GitLab and EC2 for automated updates, enhancing operational efficiency and reliability.

Associate Consultant

Jan 2023 - Jul 2023

Centelon | Bangalore

Project Title: AI Driven Document Processing and Image Labelling

Technology: YoloV5, PyTesseract, Python, Regex, Labelling, Makesense.ai

Description: Developed YoloV5 Models for Image Labelling, Specifically for Identity and non-Identity cards, Training and Testing Models on real-time data to ensure high-performance data. Created an Address Parser using Python to process Australian Addresses, breaking them into components such as Street, City, State, Country and Pincode. Additionally, Implemented an Invoice Parsing System leveraging PyTesseract and Regex to accurately extract and categorize entities like Invoice Number, Invoice Date, Amount, Sender Address and Receiver Address.

Associate Consultant

May 2022 - Dec 2022

Centelon | Bangalore

Project Title: Invoice Coordination and Product Matching at Alkem Laboratories.

Technology: Python, Excel.

Description: Managed Monthly Invoice Coordination at Alkem Pharmaceuticals, utilizing Python for accurate data extraction and handling of rejected invoices. Maintained detailed excel records of Coordinate Fixes and conducted Product Matching to ensure Precision in Medicine Name Identification during Production Processes.